Building a unified science of sensorimotor control with Densely Observable Measurment Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

> The Mission of the **FreeMoCap Foundation X-Lab (FMC-X)** is to build a novel platform technology — the **Dense Observation Measuring Environment (Dome)** — an instrument that fuses heterogeneous multi-modal instrumentation into a single **Dense Observation** representing a a calibrated, synchronized, metrologically traceable, uncertainty-tagged record of the *sensory, motor, neural, and environmental* factors underpinning goal-directed behavior of biological and artificial agents in real-world and naturalistic environments to reshape the fields of perceptomotor neuroscience, muscoloskeletal biomechanics, and mobile robotics.

Imagine a person jogging across a rocky field, and imagine everything you might measure about her. The kinematics of her body, through camera- or IMU-based motion capture. The kinetic forces at the boundary between her foot and the ground, through force plates or pressure-sensing insoles. The activity of the muscles that generated them, through electromyography. The direction of her gaze, through a mobile eye tracker; the terrain she is looking at, through photogrammetric reconstruction; the image cast upon her retina, by composing the two[1,2]. The neural activity coordinating all of it, through mobile electroencephalography — or, in a model organism, through chronic electrophysiology at cellular resolution.

Every one of those measurements has been made. **None of them have ever been made together.**

This is not for want of trying. Mobile Brain/Body Imaging[3] has pursued synchronous brain, body, and environment capture for over fifteen years, and it remains **a methodology rather than a platform**: heroic laboratories assemble it by hand, per study, out of instruments that were never built to compose. Each instrument was designed inside a discipline, to a tolerance that discipline cares about, on a clock that discipline trusts, exporting a format that discipline reads.

Ask a visual neuroscientist why the animal is head-fixed before a screen, and she will tell you it is the only way to know what the retina received. Ask a motor neuroscientist why the reach is planar and the arm is in a manipulandum, and he will tell you it is the only way to know the forces. Ask a biomechanist why the walking is steady-state on a treadmill, and she will tell you it is the only way to close the loop on the kinetics. Each is describing the same constraint from a different side: **the experiment is shaped by the instrument**. Each field has retreated to the largest region of behavior over which its own measurement stays valid, and those regions do not overlap. What lies outside all of them — an agent pursuing a goal in an unconstrained environment, using vision to place its feet — is not a niche at the edge of these fields. It is the ordinary condition of behavior, and it is the one condition none of them can measure.

Composing these instruments is not an integration exercise to be solved once by a graduate student. It is a problem in Metrology: calibration across modalities, synchronization across clocks, and the propagation of uncertainty from transducer to derived value[4,5]. It must be

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solved by the instrument itself, permanently, for everyone who uses it. And some of the constituent instruments do not exist at any price.

1.1. The New Observable: the Dense Observation

Science advances when a new **unit of observation** becomes available. The cell was not discovered by thinking harder about tissue; a lens made it observable, and a field organized itself around what the lens revealed. The study of behavior has no such unit. It has *variables* — a joint angle, a fixation, a spike train, a ground reaction force — each a projection of the agent/environment interaction onto the measurement axis of a single discipline. The projections are precise. They do not reassemble.

We define a **Dense Observation** as a single record in which the perceptual, motor, neural, and environmental state of a behaving agent are not separate datasets to be aligned after the fact, but **one measurement**: overlapping heterogeneous transducers sampling different aspects of the same whole, redundantly and at different scales, unified on a common spatial and temporal frame, with an unbroken chain of calibration behind every value and a quantified uncertainty attached to it. Under this definition gaze and gait are not two experiments to be correlated afterward; they are two faces of one observed object.

The Dense Observation is the platform technology. The instruments below are the means of producing one.

1.2. The Instrument: Dense Observation Measuring Environment (Dome)

We will develop three cross-validating variants of the Dome platform, which together form a single distributed instrument along three axes — **precision, portability, and biological depth**:

- **Dome-L** — the flagship instrumented volume, in the greater Boston area. A calibrated, synchronized room in which an entire behaving human is captured across every modality at once: markerless multi-camera motion capture, force plates, gaze, EMG, physiology, mobile EEG, and a manipulable environment (LED floor and wall panels, VR/AR, variable terrain and substrate). Large enough to fully enclose the smaller variants, which is what makes it a metrological reference rather than merely a large room. **Precision.**
- **Dome-S** — the extension of the commodity-camera capture volumes already built and disseminated through the FreeMoCap Project. Representative of the Dome a research lab or a classroom would build. **Reach.**
- **Dome-Mobile** — a wearable, self-contained Dome producing Dense Observations in unconstrained indoor and outdoor environments, continuing the lineage of the PI's gaze/gait and retinal optic flow work[1,2,6,7]. **Portability.**

The variants are not three products; they are one instrument, because **a record from a classroom kit and a record from the flagship differ in uncertainty, not in kind.** This is a claim we can make only because it is earned: Dome-Mobile is calibrated *inside* Dome-L before it is carried outdoors, establishing an unbroken traceability chain from the flagship's reference measurements to a rig on a runner's back[8,9]. The same chain extends to our collaborating animal-model laboratories, where the invasive neural modalities that cannot be applied to humans hydrate the parts of the Dense Observation we cannot reach on our own. **Biological depth.**

1.3. What We Must Build

We will buy what exists and build what does not. The clearest must-build is the eye tracker. Commercial video trackers plateau near one degree of gaze error, and **whole degrees of**

freedom of the eye are unmeasured by any instrument at any price — notably ocular torsion and the shape of the lens. Torsion was historically set aside on the authority of Listing's law, but Listing's law changes under the vestibulo-ocular reflex, which is active essentially continuously during natural locomotion. Veridical reconstruction of the retinal image of a moving agent *requires* it[1].

The physics of reaching these degrees of freedom is known and unpackaged: ocular torsion from iris texture, and accommodation from higher-order Purkinje reflections, now demonstrated at 500 Hz binocularly on commodity hardware. **This is a camera-quality bet, not a physics bet.** Smartphones proved that tiny, fast, high-resolution, inexpensive cameras exist; eye trackers never adopted them. We have already built what are plausibly the world's best eye trackers for ferrets and mice — the ferret rig integrates a three-camera skull-mounted tracker with full-body motion capture, binocular gaze, world cameras, and augmented-reality display, all calibrated as one system. **It is a working animal-scale instance of the flagship instrument.**

Three further instruments do not exist and must be built:

- **A programmatically reconfigurable capture volume.** Moving cameras is the friction that limits what an experiment can ask. A linked array of centrally controlled mounts governing extrinsics and intrinsics turns a capture volume into an addressable resource: designate a region, and the array configures itself for maximal coverage.
- **Hybrid outside-in / inside-out kinematics.** Camera-based capture yields estimates that are accurate but imprecise; IMU-based capture yields estimates that are precise but drift. Their principled fusion should exceed either, and Dome-Mobile nested inside Dome-L is the apparatus that develops and validates it.
- **A coordinated drone swarm.** Camera-carrying drones that follow an agent outdoors, avoiding obstacles and maintaining view, whose fused telemetry grounds the wearable rig's inertial drift and reconstructs the terrain the agent is traversing.

Every one of these deliverables terminates in a number: reprojection error, joint-center expanded uncertainty at $k = 2$, gaze-in-world angular uncertainty, inter-sensor synchronization jitter against a traceable clock. **The instrument is engineered so that every claim it makes is falsifiable.** That same discipline is what makes a Dense Observation machine-learning-grade — uncertainty-annotated at every sample, geometrically self-consistent across views — an instrument built from first principles for next-generation AI training pipelines[10,11].

1.4. Why This Is Unmet

These instruments do not exist because no existing structure is built to produce them. Serious, well-funded groups are building *pieces* and each is confined to its piece by its own incentives: the institutes build closed, single-species pipelines; the archives store finished, single-modality datasets[12]; the commercial markerless systems sell kinematics — and only kinematics — into verticals; the standards bodies retrofit file formats onto instruments that have already been built. The open, multimodal, calibrated, traceable capture instrument falls between every chair.

Conway's Law holds that organizations produce systems mirroring their own communication structures[13]. An enterprise of departments, journals, and funding programs named after specialized domains of inquiry will produce instruments that measure single domains, evaluated by reviewers who can only score work that sits inside one. It will not produce the long, unglamorous, cumulative labor of metrology — of turning a proof of concept into a Tool that others can trust. Tools should be built by masters and given to students, not the reverse. Building this instrument requires an organization designed to build it: full-time, autonomous, owning its own intellectual property, and funded on a horizon measured in years rather than grant cycles.

1.5. Vision

In seven years, the jogger crossing that rocky field is a routine measurement. Her kinematics, her forces, her muscle activity, her gaze, the terrain beneath her, and the image on her retina are recorded as one object, with a defensible uncertainty on every value, by an instrument that a graduate student can operate. The same record is produced, at lower coverage and higher uncertainty but not in a different kind, by a classroom in Ohio and by a ferret hunting in a laboratory in *[collaborator institution]* — and the three can be compared, because every value in each of them is traceable to the same references.

Because the record is a scaffold rather than a fixed schema, it improves in arrears: once torsion is measured jointly with the degrees of freedom that older instruments captured, it can be estimated backward into recordings made before any instrument could measure it. **The archive appreciates.**

We will build $100 three-camera kits for students, mobile eye trackers that measure degrees of freedom invisible to current technology, field-deployable gait laboratories, and wearable research environments — all of them interoperable, all of them producing the same kind of record. Heterogeneous arrays of bleeding-edge and made-to-spec instruments built into vast industrial buildings: **the opposite of a space telescope, its lenses pointing inward to the place where the body meets the world.**

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Figure 1: Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions.

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2. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - OpenCap works, but only for iPhones and requires a cloud upload, and caters exclusively to clinical biomechanics
    - SIMPL/Meshcapade - Has built past empirically validity (3d from 2d) pipelines not truth preserving. Building increasingly more abstract and less tracable data channels on top of the same 2d data we've had for a while
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehaioral Data into those sinulation/RL training enviornments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - [1,2,6]
  - etc
  - Ferret work
    -

3. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

3.1. Timeline

Phase	Name	Description
1	**Organize**	Set up Organization (Phase 0) — Solidify release of FreeMoCap v2
2	**Architect**	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	**Execute**	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	**Stabilize**	Establish OODA loop within expected standard operations
5	**Expand**	Once stabilized, expand key performance metrics
6	**Align**	Focus on building deeply integrated community of users and contributors (Transition BDFL ⟶ BDFaSPoT)
7	**Establish**	Focus on long term stability and healthy growth
8	**Maintain**	Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

4. Senior/Key Personnel Qualifications

4.1. Jonathan Matthis, PhD [Principle Investigator]

4.2. EI

4.3. AC,PhD

4.4. RR, CPA or w/e

Name / Role	Qualifications
Jonathan Matthis, PhD President	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics[1,2,6]
NR, CEO	• Expertise in technically and software dev • Previous CEO XP, founded and ran own buisiness • CEO helps PI apply his specific expertise to the core tasks of the XLab
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design and brain imaging • Worked with PI on core FMC dev and validation
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship (founded own Bookkeeping coop) • Taught Finance Theory at university level
KM Project Manager (?)	• Key driver of[2] • Worked with PI on core tech[1] • Expertise in key technologies
MN Project Manager (?)	• Expertise in clinical biomechanics, • Medical systems design, • Mechatronics for clinical tools, • Multi-modal motion capture lab management

Table 2: Senior/Key Personnel Qualifications

5. Team Capabilities Statement

5.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

5.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

5.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosenthaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

5.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

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